Long-Context Compression & Linear-Attention Memory

Research progress — Papers A / B / C

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RCA Foundation Model / Long-Context Memory

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All artifacts hyperlinked to github.com/BDeMo/research-notes (blue = clickable).

Outline / Index

1. Background: the long-context memory wall

Problem. Softmax attention keeps a *growing* KV cache: memory $\propto$ context length L. At 100k+ tokens this is the bottleneck for both cost and quality.

- **Two escape routes the field takes:**
  1. *Shrink the cache* — KV-eviction (kvzip, knorm, SnapKV…), prompt compression (LLMLingua), token merging (ToMe), retrieval (RAG).
  2. *Remove the cache* — **linear attention** / SSM (Mamba, RWKV, **Gated DeltaNet**): a *fixed-size recurrent state*, $O(L)$ time, $O(1)$ memory.
- **Our base models** live in both worlds: Qwen3 (quadratic) and **Qwen3.5/3.6** (GDN linear-attention 3:1 hybrid).

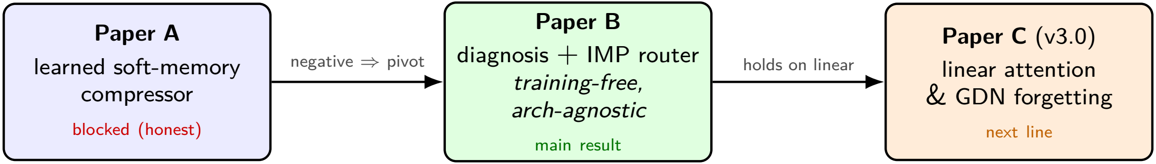
Guiding question

On a *frozen* base, how do we feed / hold the *right* context so long-context accuracy survives — across architectures?

2. One thesis, three papers (the mainline)

Thesis

Long-context accuracy is a **keep-the-answer-span** problem. No fixed compressor wins everywhere; the right mechanism is **input-adaptive** and, ideally, **training-free & architecture-agnostic**.



A learned compressor didn't beat commodity baselines (A) ⇒ we shifted to *diagnosing* what actually works and built a training-free router (B) ⇒ the diagnosis holds on cache-free linear models, opening the linear-attention line (C).

3. Paper B — the diagnosis (model-invariant)

Systematic full-test map across 9 model families (1.7B–14B, Qwen2.5/3/3.5, GLM-4, Ministral, Llama; quad + linear):

- **No universal compressor** — the winner flips by task (retrieval / extractive / MC / reasoning).
- **"More context can hurt"** — on literary-MC *full* $<$ blind guessing (QuALITY full 7.2 $<$ 17.9); compression *denoises*.
- **Retrieval $\gg$ full** on ultra-long (∞Bench) — selecting evidence beats reading everything.
- **Model-invariant**: the crossover & denoising hold on *every* family, incl. cache-free linear GDN.

This diagnosis — not a single number — is Paper B's durable contribution. fact base (F27/F31/F43) full-test main table

4. Paper B — the method: IMP (Importance-routing Prefilter)

Training-free, input-side, architecture-agnostic prefilter on a *frozen* base:

1. Score context tokens/chunks by cheap $O(L)$ signals (lexical BM25 + query-relevance + surprisal).
2. Keep the top spans/chunks *verbatim* to a budget; drop the rest. Feed to the frozen base.
3. **Route** the granularity per input (`auto`): long/extractive → chunk-BM25 over the *full doc*; literary-MC-that-fits → token-importance spans.

- No weights trained; runs on dense *and* linear bases; no KV cache needed.
- Covers regimes that RAG (fails reasoning) and KV-eviction (can't run on linear) each only half-cover.

method spec + code path   implementation

5. Paper B — full results (Qwen3-8B, all methods × benches)

bench	full·t	rag	kvzip‡	knorm‡	tome	**auto (ours)**	best
∞Bench (>100k)	52.8	64.2	35.8	25.8	51.5	**70.3**	**ours**
RULER-NIAH	99	94	**99**	85	0	98	kvzip/ours*
squad_v2	65.9	67.2	64.9	16.5	38.9	66.9	rag≈ours
trivia_qa	70.9	72.7	71.2	40.7	62.3	**72.8**	**ours**
lb_multifieldqa	38.6	38.8	37.1	30.7	–	**39.5**	**ours**
lb_narrativeqa	14.0	13.7	11.2	9.3	13.5	**14.3**	**ours**
hotpot_qa	57.2	55.8	56.2	23.1	43.9	48.8	full
QuALITY-hard	9.2	10.5	**29.9**	29.6	11.7	12.9	**kvzip‡**
MuSR	58.9	56.7	55.6	53.3	48.9	48.9	full

·t = truncated to 16k;

‡ = needs KV cache (*cannot* run on linear). ours = `auto`. complete table (16 benches, 9 families)

6. Paper B — honest standing vs *all* baselines (not just RAG)

Where ours wins / ties

- **Wins**: ∞Bench, trivia, multifieldqa, narrativeqa.
- **Ties best**: squad, lb_hotpotqa, ms_marco, LongBench-v2; best *arch-agnostic* on RULER.
- **Only method that runs on linear** among the strong ones.

Where we honestly lose

- Literary-MC (QuALITY) $\rightarrow$ *KV-eviction* (kvzip/knorm $\approx$30) — but they need a KV cache.
- Dense reasoning (MuSR, musique) $\rightarrow$ *full* (needs everything).

Framing for the paper

Contribution = the **diagnosis** + a **regime-spanning training-free router**, *not* "beats everything". We state plainly where KV-eviction / full win. best-of-all comparison

7. Paper B — rigor: a bug we found & fixed

- **Symptom:** IMP looked -19 vs RAG on ∞Bench.
- **Root cause (case-level audit):** IMP *truncated* the context to 16k *before* selecting; RAG retrieved over the *whole* 131k-token doc. **87% of RAG's kept content came from beyond 16k — IMP was structurally blind to it.**
- **Fix:** chunk-family retrieves over the full doc (no forward needed). ∞Bench 51 $\rightarrow$ **76** — the gap *reversed* to a win.
- **Discipline installed:** every result labels `full`·`trunc16k`; per-bench truncated-vs-true-full audited; only ∞Bench materially affected.

correctness audit case-level diff code review

8. Paper B — Occam simplification → v3.0

What actually drives the wins? An ablation of the (baroque) `auto` router shows:

- The **chunk branch = full-doc BM25 retrieval** carries nearly all the wins.
- Importance signals (surprisal, query-dot) add ≈ 0; the span branch only matters in literary-MC (where it still loses to KV-eviction).

v3.0 = `IMP-lite`: one mode, one signal, one knob

Full-document lexical chunk retrieval on a frozen base — no span mode, no surprisal, no router, no training. Occam: drop every non-load-bearing part; keep the workhorse. v3.0 design + validation plan

Open question (decide empirically): is IMP-lite meaningfully $>$ RAG, or is it "RAG configured for compression + the diagnosis"?

9. Paper A — learned soft-memory: an honest negative

- **Idea:** compress context into a few learned *soft tokens* on a frozen base (+ a do-no-harm gate).
- **Findings (honest):**
  - The compressor is a **commodity** — flipping any single loss moves accuracy ± 0.03.
  - Learned soft-memory **cannot compress extractive QA** (squad $\approx$ no-context); *training-free IMP beats it on the same task*.
  - The reconstruction-based gate was a **non-effect** (AUROC $\approx$ chance).
- **Outcome:** Paper A's learned compressor is the blocker $\Rightarrow$ this *motivated* the pivot to Paper B's training-free direction.

Paper A spec   negative results

10. Paper C (v3.0) — why linear attention; the research line

Focus (next line): linear attention & its variants (Qwen3.5/3.6 GDN family; *no* Qwen3 control).

- **The tension:** a *fixed-size state* is efficient but has a structural **recall / forgetting bottleneck** (recall–throughput tradeoff; state-tracking limits).
- **Our hook (already shown):** on Qwen3.5-GDN the long-context diagnosis reproduces, but **KV-eviction can't even run** (no KV) — only input-side selection (IMP-lite/RAG) works.
- **The opening:** on linear bases the problem isn't "shrink the KV" (there is none) — it's **feed the right small context / augment the state's recall**.

Candidate spine

C-1 diagnose GDN forgetting (no training) + C-2 training-free input-side memory (IMP-lite) that KV-methods structurally can't provide. research line

11. Paper C — the landscape (delta-rule family)

model	decay	erase	write	delta
Mamba-2	scalar	—	scalar	no
Gated DeltaNet (our base)	scalar	tied scalar	tied scalar	yes
KDA / Kimi-Linear	channel-wise	tied scalar	tied scalar	yes
GDN-2 (NVIDIA, 2026)	channel-wise	**channel-wise**	**channel-wise**	yes

- Frontier = **how to *edit* the compressed memory** (erase vs write) without scrambling associations.
- Field findings: the *delta rule* dominates recall; **GDN-2's erase gate carries most of its gain** (selective forgetting of stale keys is the lever); channel-wise helps long-range but can hurt precise recall.

reference collection (11 categories, arXiv)

12. Paper C — GDN forgetting: exploration & modules

Training-free anti-forgetting modules tested on Qwen3.5-GDN (real benches):

bench	full	imp-lite	rel-last	replay	rag
∞Bench	55	76	75	**78**	65
lb_multifieldqa	38.6	42.1	40.7	**44.0**	41.1
babilong-qa1	65	**68**	62	61	54

- **Input-side compression rescues GDN on ultra-long** (∞Bench 55$\rightarrow$76) — don't overflow the fixed state.
- Recency-reorder modules (rel-last / replay) = *marginal, task-dependent* $\Rightarrow$ Occam: imp-lite default.
- Honesty: our synthetic single-needle probe was *inconclusive* (task too easy / a probe bug we caught); the trustworthy evidence is the *real-bench* grid above. 128k probe OOMs (GDN kernel).

ablation design   module results (+ probe caveats)

13. Development logic & timeline

1. **Paper A** — learned soft-memory compressor $\rightarrow$ honest negatives (commodity compressor, gate a non-effect).
2. **Pivot** — stop building a compressor; *diagnose* what works on a frozen base.
3. **Paper B** — full-test diagnosis (model-invariant) + IMP router; iterated token$\rightarrow$span$\rightarrow$+lexical$\rightarrow$5-mode$\rightarrow$`auto`$\rightarrow$full-doc.
4. **Rigor pass** — case-level audit found the truncation bug (F44), fixed it, re-ran the corrected main table, best-of-all-baselines, code review.
5. **Occam** — ablate to `IMP-lite` (v3.0); own the honest scope.
6. **Paper C** — the diagnosis reproduces on linear GDN where KV-methods can't run $\rightarrow$ new line on linear-attention forgetting.

14. Cross-cutting insights & next steps

Insights

- Long-context = **keep-the-gold-span**; *gold-recall predicts accuracy*.
- **No universal compressor**; the right granularity is *input-dependent*; retrieval can *denoise above full*.
- Diagnosis is **model-invariant**; KV-eviction owns literary-MC but **can't run on linear** — that gap is ours.

Next steps

- **Paper B:** run the IMP-lite Occam ablation; finalize v3.0; decide the RAG-delta claim.
- **Paper C:** secure the in-band **Qwen3.6-27B** (hybrid) checkpoint; RULER access (HF); diagnose GDN forgetting on real long-context; test input-side vs erase-gate-retrofit fixes.

15. Artifact index (all links to GitHub)

Paper B consolidated report · complete results · main table · fact base · best-of-all correctness audit · code review · v3.0 method · 5-day summary

Paper A spec · negatives

Paper C overview · research line · references · GDN forgetting design · module results

Repo: github.com/BDeMo/research-notes